

12. Conditional expectation

def. If X and Y are jointly discrete random variables, then their conditional distribution is determined by the weight function

$$P_{X|Y}(x|y) = P(X=x|Y=y).$$

If X and Y are jointly absolutely continuous, then $X|Y$ has density

$$f_{X|Y}(x|y) = f(x,y)/f_Y(y).$$

def. Let X and Y be jointly discrete or jointly absolutely continuous random variables. The conditional expectation of X given $Y=y$ is

$$E[X|Y=y] = \begin{cases} \sum_i x_i P_{X|Y}(x_i|y) & \text{in the discrete case, } X \in \{x_i\}_{i \in I} \\ \int x f_{X|Y}(x|y) & \text{in the absolutely continuous case} \end{cases}$$

Example: Let $X \sim \text{Bin}(n, p)$, $Y \sim \text{Bin}(m, p)$ be independent random variables
 $E[X|X+Y=k] = ?$

Let $0 \leq k \leq m+n$, $0 \leq i \leq \min\{n, k\}$, then

$$P(X=i|X+Y=k) = \frac{P(X=i, X+Y=k)}{P(X+Y=k)} = \frac{P(X=i, Y=k-i)}{P(X+Y=k)} \quad \leftarrow \begin{array}{l} \text{they are} \\ \text{independent} \end{array}$$

$$= \frac{\binom{n}{i} p^i (1-p)^{n-i} \binom{m}{k-i} p^{k-i} (1-p)^{m-(k-i)}}{\binom{n+m}{k} p^k (1-p)^{n+m-k}} = \frac{\binom{n}{i} \binom{m}{k-i}}{\binom{n+m}{k}} \Rightarrow X|X+Y=k \sim \text{Hyp}(n+m, n, k)$$

the conditional distribution is hypergeometric

$$X+Y \sim \text{Bin}(n+m, p)$$

$$\Rightarrow E[X|X+Y=k] = \frac{n}{n+m} k$$

Note: $G(y) = E[X|Y=y]$ is a function of y . $E[X|Y] = G(Y)$ is a random variable!

Prop. Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$, then

$$E[g(X,Y)h(Y)|Y] = h(Y)E[g(X,Y)|Y]$$

proof: $G(y) = E[g(X,Y)h(Y)|Y=y] = E[g(X,y)h(y)|Y=y] = h(y)E[g(X,y)|Y=y]$

$$\Rightarrow E[g(X,Y)h(Y)|Y] = G(Y) = h(Y)E[g(X,Y)|Y] \quad \square$$

Prop. (Tower-rule)

Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a mapping, and let X, Y be random variables.

Then $E[g(X,Y)] = E[E[g(X,Y)|Y]]$

proof: We only prove it for jointly absolutely continuous X and Y .
The proof in the jointly discrete case is analogous.

$$\begin{aligned} G(y) &= E[g(X, Y) | Y=y] \\ E[E[g(X, Y) | Y]] &= E[G(Y)] = \int G(y) f_Y(y) dy = \int E[g(X, Y) | Y=y] f_Y(y) dy = \\ &\stackrel{\text{def}}{=} \iint g(x, y) f_{X|Y}(x|y) dx f_Y(y) dy = \iint g(x, y) \underbrace{f_{X|Y}(x|y) f_Y(y)}_{= f(x, y)} dx dy = E[g(X, Y)] \quad \square \end{aligned}$$

Example: A miner lost in his mine. He forgot which gate leads outside, so he has to choose one uniformly.

X = number of hours it takes for him to get out $E[X] = ?$

Let Y = the gate he chooses, then (for the first time)

$$E[X] \stackrel{\text{tower-rule}}{=} E[E[X|Y]] = \sum_{i=1}^3 E[X|Y=i] \underbrace{P(Y=i)}_{= 1/3} =$$



$$= \frac{1}{3} \cdot 3 + \frac{1}{3} E[X+7] + \frac{1}{3} E[X+5] = 1 + \frac{5+7}{3} + \frac{2}{3} E[X] \Rightarrow \underline{\underline{E[X] = 18}}$$

$Y=1$: he gets out after 3 hours of walking $\Rightarrow E[X|Y=1] = 3$
 $Y=2$: he walks for 7 hours just to come back to the same room
 $\Rightarrow E[X|Y=2] = E[X+7]$
 $Y=3$: he walks for 5 hours and then arrives to the same room
 $\Rightarrow E[X|Y=3] = E[X+5]$

def: Let X be a random variable, and let A be an event.
 $P(A|X) = E[1_A | X]$

Cor: By the tower-rule, $E[P(A|X)] = E[E[1_A | X]] = E[1_A] = P(A)$
 It is basically the law of total probability.

def: Let X and Y be random variables. We define their conditional variance as
 $D^2[X|Y] = E[(X - E[X|Y])^2 | Y]$

$$\text{Prop: } D^2[X|Y] = E[X^2|Y] - E^2[X|Y]$$

$$\text{" } D^2[X] = E[D^2[X|Y]] + D^2[E[X|Y]]$$

proof: $D^2[X|Y] = E[(X - E[X|Y])^2 | Y] = E[X^2 - 2X E[X|Y] + E^2[X|Y] | Y] =$
 $\stackrel{\text{linearity of } E}{=} E[X^2|Y] - 2E[X E[X|Y] | Y] + E[E^2[X|Y] | Y] = E[X^2|Y] - 2E^2[X|Y] + E^2[X|Y]$
 $E[X|Y]$ is a function of Y
 + prop.

$$1) D^2[X] = E[X^2] - E^2[X] = E[E[X^2|Y]] - E^2[E[X|Y]] =$$

tower rule

$$= E[E[X^2|Y]] - E[E^2[X|Y]] + E[E^2[X|Y]] - E^2[E[X|Y]] = E[D^2[X|Y]] + D^2[E[X|Y]] \quad \square$$

Thm (Steiner) Let X be a random variable, then

$$E[(X-c)^2] = D^2[X] + (c - E[X])^2 \quad \text{for } c \in \mathbb{R}.$$

That is $\min_{c \in \mathbb{R}} E[(X-c)^2] = D^2[X]$, and the minimum is attained at $c = E[X]$

proof: $E[(X-c)^2] = E[(X - E[X] + E[X] - c)^2] = E[(X - E[X])^2 + 2(X - E[X])(E[X] - c) + (E[X] - c)^2] =$
 $= E[(X - E[X])^2] + 2(E[X] - c) \underbrace{E[X - E[X]]}_{=0} + E[(E[X] - c)^2] = D^2[X] + (E[X] - c)^2 \quad \square$

Thm (Steiner) Let X, Y be random variables, and $G: \mathbb{R} \rightarrow \mathbb{R}$, then

$$E[(X - G(Y))^2 | Y] = D^2[X|Y] + (E[X|Y] - G(Y))^2$$

In particular, $\min_{G: \mathbb{R} \rightarrow \mathbb{R}} E[(X - G(Y))^2 | Y] = D^2[X|Y]$ and attained at $G(Y) = E[X|Y]$

proof: $E[(X - G(Y))^2 | Y] = E[(X - E[X|Y] + E[X|Y] - G(Y))^2 | Y] =$
 $= E[(X - E[X|Y])^2 | Y] + 2 \underbrace{E[(X - E[X|Y])(E[X|Y] - G(Y)) | Y]}_{\text{function of } Y} + E[(E[X|Y] - G(Y))^2 | Y] =$
 $\underbrace{D^2[X|Y]}_{\text{function of } Y} + 2(E[X|Y] - G(Y)) \underbrace{E[X - E[X|Y] | Y]}_{=0} + (E[X|Y] - G(Y))^2$
 $\xrightarrow{\text{prop.}} D^2[X|Y] + (E[X|Y] - G(Y))^2 \quad \square$

Prop Let X, Y and Z be random variables. Then

$$E[X + Y | Z] = E[X | Z] + E[Y | Z]$$

proof. We only prove it in the jointly absolutely continuous case.

$$H(z) := E[X + Y | Z = z] = \iint (x + y) f_{X,Y|Z}(x, y | z) dx dy =$$

$$f_{X,Y|Z}(x, y | z) = \frac{f(x, y, z)}{f_z(z)}, \quad \text{where } f_z(z) \neq 0 \text{ and } f(x, y, z) \text{ is the joint density function of } X, Y, Z$$

$$= \iint x f_{X,Y|Z}(x, y | z) dx dy + \iint y f_{X,Y|Z}(x, y | z) dx dy = \underbrace{\int x f_{X|Z}(x | z) dx}_{= E[X|Z=z]} + \underbrace{\int y f_{Y|Z}(y | z) dy}_{= E[Y|Z=z]}$$

$$= \underbrace{\int x \int \frac{f(x, y, z)}{f_z(z)} dy dx}_{= \frac{f_{X|Z}(x | z)}{f_z(z)} = f_{X|Z}(x | z)} + \underbrace{\int y \int \frac{f(x, y, z)}{f_z(z)} dx dy}_{= \frac{f_{Y|Z}(y | z)}{f_z(z)} = f_{Y|Z}(y | z)}$$

Thus, if $f_2(z) \neq 0$, then

$$\mathbb{E}[X+Y|Z=z] = H(z) = \mathbb{E}[X|Z=z] + \mathbb{E}[Y|Z=z]$$

$$\Rightarrow \mathbb{E}[X+Y|Z] = H(Z) = \mathbb{E}[X|Z] + \mathbb{E}[Y|Z]$$

□

Example: Suppose that we send an $S \sim N(\mu, \sigma^2)$ signal on a network.

The network is noisy, so the received signal R has conditional distribution

$$R|S \sim N(S, 1)$$

What is our best guess for S if we receive $R=r$?

According to Steiner's theorem our best guess is $\mathbb{E}[S|R=r]$.
(also the best in conditional expectation simply)

$$f_{S|R}(s|r) = \frac{f_{S,R}(s,r)}{f_R(r)} = \frac{f_{R|S}(r|s) f_S(s)}{f_R(r)} = \frac{1}{f_R(r)} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(r-s)^2}{2}} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(s-\mu)^2}{2\sigma^2}}$$

In the exponent we have

$$\begin{aligned} -\frac{1}{2} \left((r-s)^2 + \frac{(s-\mu)^2}{\sigma^2} \right) &= -\frac{1}{2} \left(r^2 - 2rs + s^2 + \frac{s^2}{\sigma^2} - \frac{2s\mu}{\sigma^2} + \frac{\mu^2}{\sigma^2} \right) = -\frac{1}{2} \left(\frac{1+\sigma^2}{\sigma^2} s^2 - 2s \left(r + \frac{\mu}{\sigma^2} \right) + r^2 + \frac{\mu^2}{\sigma^2} \right) \\ &= -\frac{1}{2} \frac{1+\sigma^2}{\sigma^2} \left(s^2 - 2s \left(r + \frac{\mu}{\sigma^2} \right) \frac{\sigma^2}{1+\sigma^2} + \left(r + \frac{\mu}{\sigma^2} \right)^2 \frac{\sigma^4}{(1+\sigma^2)^2} \right) - \frac{1}{2} \left(r^2 + \frac{\mu^2}{\sigma^2} \right) \\ &= -\frac{\left(s - \frac{\mu}{1+\sigma^2} - \frac{r\sigma^2}{1+\sigma^2} \right)^2}{2 \frac{\sigma^2}{1+\sigma^2}} + \frac{1}{2} \left(\left(r + \frac{\mu}{\sigma^2} \right)^2 \frac{\sigma^2}{1+\sigma^2} + r^2 - \frac{\mu^2}{\sigma^2} \right) \\ &= -\frac{\left(s - \frac{\mu}{1+\sigma^2} - \frac{r\sigma^2}{1+\sigma^2} \right)^2}{2 \frac{\sigma^2}{1+\sigma^2}} - \frac{\left(s - \frac{\mu}{1+\sigma^2} - \frac{r\sigma^2}{1+\sigma^2} \right)^2}{2 \frac{\sigma^2}{1+\sigma^2}} + \frac{(r-\mu)^2}{2(1+\sigma^2)} \\ &= r^2 \left(\frac{\sigma^2}{1+\sigma^2} - 1 \right) + \frac{2r\mu}{1+\sigma^2} + \frac{\mu^2}{\sigma^2} \left(\frac{1}{1+\sigma^2} - 1 \right) = -\left(\frac{r^2}{1+\sigma^2} - \frac{2r\mu}{1+\sigma^2} + \frac{\mu^2}{1+\sigma^2} \right) = -\left(\frac{r}{\sqrt{1+\sigma^2}} - \frac{\mu}{\sqrt{1+\sigma^2}} \right)^2 \end{aligned}$$

$$\Rightarrow f_{S|R}(s|r) = \frac{1}{\sqrt{2\pi}\sigma} \frac{1}{f_R(r)} e^{-\frac{1}{2} \frac{(r-\mu)^2}{1+\sigma^2}} e^{-\frac{\left(s - \frac{\mu}{1+\sigma^2} - \frac{r\sigma^2}{1+\sigma^2} \right)^2}{2 \frac{\sigma^2}{1+\sigma^2}}}$$

Since $\int f_{S|R}(s|r) ds = 1$ we don't need to calculate $f_R(r)$ to tell that

$$\begin{aligned} f_{S|R}(s|r) &= \frac{1}{\sqrt{2\pi} \frac{\sigma}{\sqrt{1+\sigma^2}}} e^{-\frac{\left(s - \frac{\mu}{1+\sigma^2} - \frac{r\sigma^2}{1+\sigma^2} \right)^2}{2 \frac{\sigma^2}{1+\sigma^2}}} \\ &\Rightarrow S|R=r \sim N\left(\frac{\mu}{1+\sigma^2} + \frac{r\sigma^2}{1+\sigma^2}, \frac{\sigma^2}{1+\sigma^2} \right) \\ &\Rightarrow \mathbb{E}[S|R=r] = \frac{\mu}{1+\sigma^2} + \frac{r\sigma^2}{1+\sigma^2} \end{aligned}$$

Thus the best guess is a weighted average of the received signal r and the average signal μ . If the variance of the sent signal S is big, that is the noise is smaller than σ , then r is the better guess. If σ is small, which means that the noise is relatively big, then μ is better.

Example: (Sum of randomly many terms)

In a supermarket N people are shopping each day.

X_i = the money that customer i spends, Y = daily income of this supermarket

$$\mathbb{E}[Y] = ? \quad \mathbb{D}^2[Y] = ?$$

Suppose that X_i are iid. with $\mathbb{E}[X_i] = \mu$ and $\mathbb{D}^2[X_i] = \sigma^2$, and all of them are independent of N .

Since $Y = \sum_{i=1}^N X_i$, we have

$$\begin{aligned} \underline{\underline{E[Y]}} &= \underline{\underline{E\left[\sum_{i=1}^N X_i\right]}} = \underline{\underline{E\left[E\left[\sum_{i=1}^N X_i \mid N\right]\right]}} = \sum_{k=0}^{\infty} E\left[\sum_{i=1}^k X_i\right] \cdot P(N=k) = \\ &\quad \text{tower-rule} \\ &= \sum_{k=0}^{\infty} \sum_{i=1}^k E[X_i] P(N=k) = \sum_{k=0}^{\infty} k \cdot \mu \cdot P(N=k) = \underline{\underline{\mu E[N]}} \end{aligned}$$

$$\underline{\underline{D^2[Y]}} = \underline{\underline{E\left[D^2\left[\sum_{i=1}^N X_i \mid N\right]\right]}} + \underline{\underline{D^2\left[E\left[\sum_{i=1}^N X_i \mid N\right]\right]}} = \underline{\underline{\sigma^2 E[N] + \mu^2 D^2[N]}}$$

X_i are iid. $\Rightarrow \sum_{i=1}^N D^2[X_i] = \sigma^2 N$
and indep. of N

Example: Let $\tau_1, \tau_2, \dots$ be iid random variables with distribution $\text{Exp}(\lambda)$.

Let $t \in \mathbb{R}_+$ and define $N_t = \max_{n \in \mathbb{N}} \{n \geq 1, \tau_1 + \dots + \tau_n \leq t\}$

What is the distribution of N_t ?

$$P(N_t = k) = \underline{\underline{P(\tau_1 + \dots + \tau_k \leq t < \tau_1 + \dots + \tau_{k+1})}} = E[P(\tau_k \leq t < \tau_k + \tau_{k+1} \mid \tau_k)] =$$

$$= E[\underbrace{P(\tau_k \leq t < \tau_k + \tau_{k+1})}_{\tau_k = \sum_{i=1}^k \tau_i}] = E[\mathbb{1}_{\{\tau_k \leq t\}} P(t - \tau_k < \tau_{k+1} \mid \tau_k)] = E[\mathbb{1}_{\{\tau_k \leq t\}} e^{-\lambda(t - \tau_k)}] =$$

$\tau_{k+1} \sim \text{Exp}(\lambda)$
and indep. of $\tau_1, \dots, \tau_k$

$$= \int_{\mathbb{R}^k} \int \mathbb{1}_{\{t_1 + \dots + t_k \leq t\}} e^{-\lambda(t - t_1 - \dots - t_k)} \cdot \underbrace{\lambda^k e^{-\lambda(t_1 + \dots + t_k)}}_{P(\tau_1 = t_1) \dots P(\tau_k = t_k)} dt_1 \dots dt_k =$$

$$= e^{-\lambda t} \lambda^k \int_{\mathbb{R}^k} \int \mathbb{1}_{\{t_1 + \dots + t_k \leq t\}} dt_1 \dots dt_k = e^{-\lambda t} \lambda^k \underbrace{\int_0^t \int_0^{t-t_1} \int_0^{t-t_1-t_2} \dots \int_0^{t-t_1-\dots-t_{k-1}} 1 dt_k \dots dt_1}_{= \frac{t^k}{k!}, \text{ volume of the } k\text{-dimensional simplex}}$$

[We can prove it by mathematical induction:

• $k=1$: $\int_0^t 1 dt_1 = t$

• Suppose that it holds for $k-1$.

• For k : $t, t-t_1, t-t_1-t_2, \dots, t-t_1-\dots-t_{k-1}$

$$\left[\int_0^t \int_0^{t-t_1} \int_0^{t-t_1-t_2} \dots \int_0^{t-t_1-\dots-t_{k-1}} 1 dt_k \dots dt_2 dt_1 = \left[-\frac{(t-t_1)^k}{k!} \right]_0^t = \frac{t^k}{k!} \right]$$

$= \frac{(t-t_1)^{k-1}}{(k-1)!}$ by assumption

$$\Rightarrow P(N_t = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!} \Rightarrow N_t \sim \text{Poi}(\lambda t)$$

This is the Poisson process with rate λ .